

Exercise 7:

a) Find the correlation coef. (r)
In a simple linear regression
 $r^2 = R^2$

$$r = \frac{\sum (n - \bar{n})(y - \bar{y})}{\sqrt{\sum (n - \bar{n})^2 \sum (y - \bar{y})^2}} \quad \bar{n} =$$

$$\underline{\underline{r = -0.8224}}$$

b) Calculate point estimation of the slope ($\hat{\beta}_1$) and intercept ($\hat{\beta}_0$)

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (n - \bar{n})(y - \bar{y})}{\sum (n - \bar{n})^2} = -1533.634$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 18357.74$$

x : age y : cost

c) Estimate the cost of a 5 y. old Car

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \quad x=5, \quad \hat{\beta}_0 = 18357.74, \quad \hat{\beta}_1 = -1533.634$$

$$\hat{y} = 18357.74 - 1533.634 \times 5$$

$$\boxed{\hat{y} = 10689.56}$$

$$\alpha = 0.1$$

$$H_0: \beta_1 = 0 \quad H_1: \beta_1 \neq 0$$

$$t_{\text{obs}} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \sim T_{n-p-1}, 1-\frac{\alpha}{2}$$

$$n=14, p=1$$

$$\boxed{t_{\text{obs}} = -5.008}$$

$$t_{\text{stat}} = t_{12, 1-\frac{0.1}{2}}$$

$$= t_{12, 0.95}$$

$$= 1.782$$

$$|t_{\text{obs}}| > t_{\text{stat}}$$

$$5.008 > 1.782$$

we reject H_0 hence β_1 est. significant.

$$\rightarrow H_0: \text{SOFCA} < 1$$

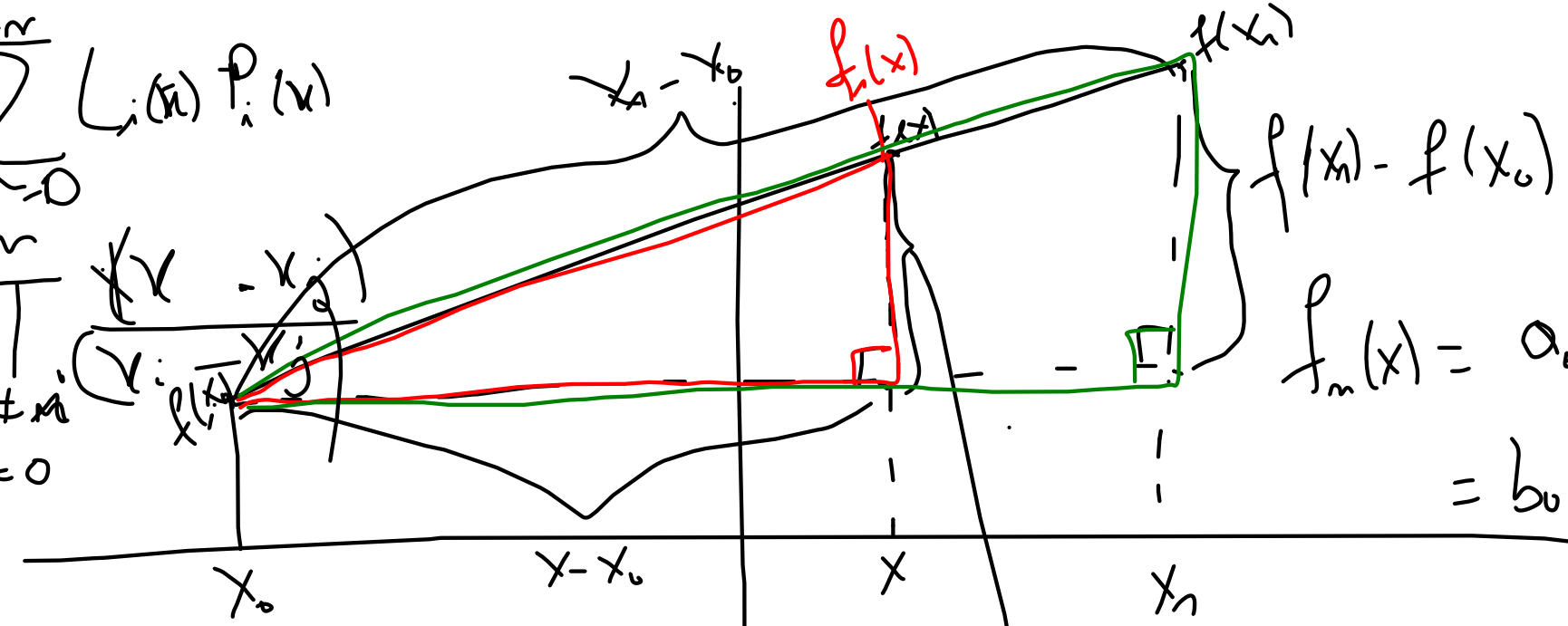
$$\rightarrow H_1: \underline{\underline{\text{SOFCA} \geq 1}}$$

x	y

$$y = \underset{\substack{\uparrow \\ \text{Intercept}}}{\beta_0} + \underset{\substack{\uparrow \\ \text{slope}}}{\beta_1} x$$

$$L f(x) = \sum_{i=0}^{\infty} L_i(x) p_i(x)$$

$$L_i(x) \sim \prod_{j=0, j \neq i}^{\infty} \frac{x - x_j}{x_i - x_j}$$



$$f_n(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

$$= b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_n) + \dots + b_m(x-x_0)\dots(x-x_n)$$

$$\frac{f_1(x) - f(x_0)}{x - x_0} = \frac{f(x_n) - f(x_0)}{x_n - x_0}$$

$$f_1(x) - f(x_0)$$

$$f_1(x) = \left(\frac{f(x_n) - f(x_0)}{x_n - x_0} \right) (x - x_0) + f(x_0)$$

a
b
-
0
-
-
f

